

FedShield-ID

Privacy-First Identity Trust Platform for Banking Networks

Hackathon Pitch Deck

Topic: Cybersecurity & Privacy-Preserving AI in Financial Tech

Slide 1: Problem & Solution

The Challenge in Banking Systems

- **Identity Silos & Centralization:** Legacy architectures pool raw customer PII. This violates banking regulations and creates massive data breach liabilities.
- **Synthetic ID & Account Takeovers:** Fraudsters bypass static onboarding checks and hijack account recovery channels (e.g. SIM swaps) before transactions occur.
- **Robotic Sweeps:** Credential stuffing attacks utilize automated bot scripts, mimicking standard logins but clicking in sub-millisecond rates.

The FedShield-ID Approach

- **Decentralized Machine Learning:** Models are trained locally at individual banking nodes without data pooling. Global models are merged via secure FedAvg federated rounds.
- **Continuous Multi-Dimension Scoring:** Aggregates 10 scoring parameters (reputation, location, biometrics, insider activity) into an active trust score (0-100).
- **Risk-Based Step-Up Challenges:** Triggers dynamic security checks (like Biometric Face ID) only when scoring parameters indicate risk anomalies.

Slide 2: Technology & Innovation

System Architecture & Tech Stack

- **Backend Framework:** FastAPI microservices coupled with SQLAlchemy ORM running on SQLite/PostgreSQL databases.
- **Machine Learning Core:** Scikit-Learn (SGD & Random Forest) for local classification, coupled with linear SHAP explainer code for XAI.
- **Interactive SOC Interface:** React 18 frontend built with Tailwind CSS, SVG linkage charts (Knowledge Graph), and live streaming transaction ledgers.

Core Cryptography & Telemetry

- **Post-Quantum KEM Tunnels:** CRYSTALS-Kyber-768 key encapsulation matrix polynomial equations to secure node-to-node communications.
- **Differential Privacy (DP):** Laplacian noise addition to local model parameters controlled by a privacy budget (epsilon, ϵ) to prevent database leaking.
- **Behavioral Biometrics:** Live tracking of keystrokes/minute and mouse jitter standard deviation to stop headless browsers.

Slide 3: Scalability & Deployment

Ecosystem Integration & Feasibility

- **Asynchronous Telemetry Ingestion:** Feeds telemetry data in background threads without interfering with main database queries.
- **Sub-Millisecond Overhead:** Crystals-Kyber simulated KEM and local SHAP mathematical computations run in microsecond scales, preserving transaction velocity.
- **Containerized Orchestration:** Docker-compose setup makes local deployment reproducible across test nodes and cloud platforms.

Platform Impact & Scale

- **Privacy compliance:** Meets strict international privacy regulations (GDPR, RBI banking privacy criteria) by removing the centralized data lake.
- **Mitigation Playbooks:** Incident response times are reduced using automated GenAI security briefings for SOC operators.
- **Money Laundering Prevention:** SVG Knowledge Graph identifies account clusters linked to shared compromised devices or VPN nodes.

Slide 4: Team & Execution Plan

Team Roles & Structure

- **AI/ML Architect (Mohd. Amaan Hamid):** Handles federated averages, differential privacy, local ML training models, and SHAP explainability variables.
- **Security Engineering:** Configures simulated crystals-kyber KEM, benchmark evaluations, and insider threat audit parameters.
- **Web Dashboard Engineer:** Designs React frontend panels, responsive Tailwind styling, and live-streaming ledgers.

Development Roadmap & Milestones

- **Milestone 1 (Current):** Completed local prototype running 3 banks with live simulator, SVG knowledge graph, and inline XAI drawer.
- **Milestone 2 (Q3):** Deploy independent containerized bank nodes in clouds and configure multi-channel secure gRPC threads.
- **Milestone 3 (Q4):** Integrate production liboqs library files and package browser-level keyboard/mouse biometrics SDK.